

ReadMe Document

This ReadMe document will help you with a quick navigation through the contents of the Pre-Reading material, Preview videos, and Individual Labs for the upcoming session.

Outcomes

After completion of this week's sessions, & activity participants will be able to :

1. Understand how and what CNN learns to identify a particular image & the concept of Transfer learning.
2. Understand the concept of global minima and better optimizers than Gradient Descent.
3. Understand & implement RNN & LSTM.
4. Understand different applications of speech processing and techniques used for this viz. feature extraction techniques, different modeling methods, and finally advanced AI & ML applications in Speech Systems.

Lecture Topics

1. Appreciating CNN & Advances in Training
2. LSTM & GRU
3. Foundations in Speech Processing & ML in Speech
4. Review Sessions

Preview Videos

The Preview Videos Link for the respective sessions are available a week in advance (i.e. every Monday), to help you understand the concepts that will be covered in the upcoming lectures. There could be one or more videos depending on the lecture details. The video/videos should be viewed before attending the lecture for better understanding.

There are 4 preview videos for this session and they should be viewed as given below:

1. Appreciating CNNs: This is a 4-minute video. From this video, you will understand 1. Dot product 2. Similarity between the images using dot product between vectors 3. Correlation between two images 4. How linear classification works for cats and dogs dataset.
2. Advance in Training: This is a 3-minute video. This video talks about Escaping Local minima and identifying the best weights to minimize the loss. Convex vs non-convex and local vs Global minima
3. LSTM and GRU: This is a 8 mins video. From this video, you will understand the basics of LSTM and GRU.
4. This session consists of one preview video - "Speech Recognition" : This is a 3 minute 13 sec video. In this video, you will be introduced to speech recognition and its different applications.

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Reference Material

The Reference Material is given on (1) RNN, (2) MFCC (Repeated from Module2), (3) Speech Production .. You should read the material after attending the lecture. This reference material will provide further depth into the topic.

Individual Labs

1. CNN
2. RNN, LSTM and GRU
3. Speech Processing